The dynamic balancing machine is used to find the vibrations of armature or rotor when it rotates fast and it also gives the details of unbalance condition. i.e, which the part has more weight or less weight in the armature.

The accurate balancing of armature is not possible in static balancing method. But in dynamic balancing method, it can be done properly. While testing, the rotor is rotated fast. If it is unbalance, the weight is added to the required to balance the rotor.

In large type machines, the armature balancing is tested by using electronic device, oscilloscope and vibration detectors. If balance is not correct, it will damage the bearings and the laminated core in the armature and the commutator becomes loose. This makes the armature to vibrate fast and produces sound. The motor starts to move from its position. If it is not in proper. bolt it with ground.

The fan kept in the motor is used to reduce the heat. The weight can be added on the fan's wing for balance. For balancing the method of adding weight is better than reducing the weight.

10.6.6 End play

The motor shaft is fixed without moving front and back position in length wise. The end play occurs while the shaft is moving more than $1/64^{th}$ part of inch is called end play.

10.6.7 End thrust

Unwanted force acting on the shaft sides of the motor or towards the one end of the motor is called end thrust.

10.6.8 Bearing

To support and noiseless rotation of shaft, bearing are used. The place where the bearing are installed are known as housing of bearing. Bearing are used in the end cover of the machine and shaft. With bearing puller, we can easily remove the bearing.

10.7 Testing of windings

For a reliable operation of electrical machines, it is important it they must be tested at regular intervals. Over to all, testing is important to know about the type of fault and to desire the cost and time required to rectify the fault.

Generally four types of tests were conducted in windings. They are,

1. Winding resistance test
2. Insulation resistance test
3. Growler test
4. Drop test

1. Winding resistance test

Winding resistance test is used for finding the short circuits within the coil between conductors and also between the coils. Low voltage DC supply should be given to the windings pay connecting ammeter in series and the voltmeter in parallel with the winding and readings are taken and tabulated.

According to ohm's law, we know that,

Winding resistance test

$$R = \frac{voltage\ applied\ to\ winding)}{current}$$

Resistance value must be calculated for each pole winding and phase winding. If the resistance value of pole or phase winding is low in value, then there is a short circuit in the particular pole or phase winding.

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